- Introduction & Problem Statement

Analysis of political data of Indian state level election for different years for analysis of metrics likes total votes polled, electors with different parties etc

- Dataset Description

The dataset contains political data of elections held for different years and different states containing data including candidate names , party names, ac_type and total votes polled and electors

- Data Cleaning Summary

- No.of rows & columns is: (73081, 11)
- Statistical data loaded with describe() function
- Checked for duplicates with duplicated().sum()
- Almost No duplicates found which is conformed with drop_duplicates() after which data is same
- Analyzed missing values to get missing count and missing percentage. There are some missing data in pc_type and cand_sex columns
- Unnecessary columns like 'partyabbre', 'pc_no','pc_type' are dropped
- None data in cand_sex filled with “unknown”
- The data of votes polled analyzed with respect to partyname with groupby() and aggregated data like sum, mean and count of totalvotespolled is extracted for each party name
- Divided the data in numerical and categorical data with select_dtypes() function
- Analyzed data to find outliers in columns like 'totvotpoll' and 'electors'
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- Key Visualizations with Explanation

- Box plot generated with seaborn library for “totvotpoll” to show outliers in total votes polled
- Kdeplot generated for “totvotpoll” as part of univariate analysis
- Bivariate analysis done for the variation of electors year wise with bar plot
- The correlation between the all columns of data are plotted with generating heat map to show correlation of data with every other data
- The 3d scatter plot with three columns data 'year', 'totvotpoll', 'electors' is generated as multivariate analysis
- The multivariate analysis also done with help of pair plot for different partyname

- Insights & Findings – Conclusion

- No.of rows & columns is: (73081, 11)
- Almost No duplicates found which is conformed with drop_duplicates() after which data is same
- Analyzed missing values to get missing count and missing percentage. There are some missing data in pc_type and cand_sex columns
- None data in cand_sex filled with “unknown” which is needed if have to analyze the data for each gender. Replacing null data with mode might fill most of data with male so it might be misleading. Please add your suggestion here
- The data of votes polled analyzed with respect to partyname with groupby() which shows the ‘AAAP’ got more votes polled following ‘A-Chik National Congress(Democratic) ‘ where ‘A S P’ got less among all
- Divided the data in numerical and categorical data with select_dtypes() function which is useful in calculating key aspects like outliers and dropping data if not necessary
- Analyzed data to find outliers in columns like 'totvotpoll' and 'electors', Around 21% of outliers found in totvotpoll and 2.079% in electors
- The kdeplot of totalvotes polled shows the data is more tend to lie towards right, so it is right skewed or positive skewed
- Bivariate analysis of electors done for year wise to show how its varying year by year which is increasing in later year except in 2010 it is dropped slightly
- By heatmap to show correlation between the columns we can conclude there good correlation between electors and year because as years progressed the electors are almost increased , where total votes polled also have just better correlation with electors and year
- The pairplot will show the relation with data and every other data which is devided for each party name with hue which is multivariate analysis